1. $D=\begin{pmatrix}10 & \\ & 5\end{pmatrix}$ $L=\begin{pmatrix}\\ -c & \end{pmatrix}$ $U=\begin{pmatrix} & -a\\ \\ \end{pmatrix}$

$B_J=D^{-1}(L+U)=\begin{pmatrix}0 & -\frac{a}{10}\\ -\frac{c}{5} & 0\end{pmatrix}$

$\det(\lambda I-B_J)=\lambda^2-\frac{ac}{50}$

若 $ac\geq 0$ $\lambda=\pm\sqrt{\frac{ac}{50}}$ $\rho(B_J)<1\Rightarrow 0\leq ac<50$

若 $ac<0$ $\lambda=\pm\sqrt{-\frac{ac}{50}}i$ $\rho(B_J)<1\Rightarrow -50<ac<0$

$\therefore -50<ac<50$ Jacobi

$B_{GS}=(D-L)^{-1}U=\begin{pmatrix}0 & -\frac{a}{10}\\ 0 & \frac{ac}{50}\end{pmatrix}$

$\det(\lambda I-B_{GS})=\lambda(\lambda-\frac{ac}{50})$ $\rho(B_{GS})<1$ $-50<ac<50$

2. (1) $x^{(k+\frac{1}{2})}=(D-L)^{-1}Ux^{(k)}+(D-L)^{-1}b$

$(D-U)x^{(k+1)}=L(D-L)^{-1}Ux^{(k)}+L(D-L)^{-1}b+b$

$x^{(k+1)}=(D-U)^{-1}L(D-L)^{-1}Ux^{(k)}+(D-U)^{-1}L(D-L)^{-1}b+(D-U)^{-1}b$

① $(D-U)^{-1}L(D-L)^{-1}+(D-U)^{-1}=(I-\tilde{U})^{-1}(I-\tilde{L})^{-1}D^{-1}$

$\Leftrightarrow (I-\tilde{U})^{-1}D^{-1}(L(D-L)^{-1}+I)=(I-\tilde{U})^{-1}(I-\tilde{L})^{-1}D^{-1}$

$\Leftarrow L(D-L)^{-1}+I=D(D-L)^{-1}$

$(D-L)(D-L)^{-1}=I$ 成立

② $(D-U)^{-1}L(D-L)^{-1}U=(I-\tilde{U})^{-1}(I-\tilde{L})^{-1}\tilde{L}\tilde{U}$

$\Leftrightarrow (I-\tilde{L})^{-1}\tilde{L}(I-\tilde{L})^{-1}\tilde{U}=(I-\tilde{U})^{-1}(I-\tilde{L})^{-1}\tilde{L}\tilde{U}$

$\Leftrightarrow \tilde{L}(I-\tilde{L})^{-1}=(I-\tilde{L})^{-1}\tilde{L}$

$\Leftrightarrow (I-\tilde{L})\tilde{L}=\tilde{L}(I-\tilde{L})$ 显然

(2) 由A正定，所以D每个元素均大于0

设λ为B的特征值

$(I-\tilde{U})^{-1}(I-\tilde{L})^{-1}\tilde{L}\tilde{U}\vec{x}=\lambda\vec{x}$

$\tilde{L}\tilde{U}\vec{x}=\lambda(I-\tilde{L}-\tilde{U}+\tilde{L}\tilde{U})\vec{x}$

$D\tilde{L}\tilde{U}\vec{x}=\lambda(A+D\tilde{L}\tilde{U})\vec{x}$

$L=D\tilde{L}=U^T=\tilde{U}^TD^T=\tilde{U}^TD$

$\lambda=\frac{x^TD\tilde{L}\tilde{U}x}{x^TAx+x^TD\tilde{L}\tilde{U}x}=\frac{x^T\tilde{U}^TD\tilde{U}x}{x^TAx+x^T\tilde{U}^TD\tilde{U}x}$

$C\sqrt{D}$即可

~~证明~~ $\tilde{U}^TD\tilde{U}$为半正定矩阵

$\therefore x^T\tilde{U}^TD\tilde{U}x\geq 0$ 而 $x^TAx>0$, $\therefore \lambda\geq 0$

(3) $\lambda=1-\frac{x^TAx}{x^TAx+x^T\tilde{U}^TD\tilde{U}x}$

$x^TAx>0$ $x^T\tilde{U}^TD\tilde{U}x\geq 0$

$\therefore \lambda<1$ 即$\rho(B)<1$

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3. $B_{SOR}^{\omega} = (D-\omega L)^{-1}[(1-\omega)D+\omega U]$ 设特征值为 λ

$\det(\lambda I - B_{SOR}^{\omega}) = 0$

$\det((D-\omega L)^{-1}) \det(\lambda(D-\omega L) - [(1-\omega)D+\omega U]) = 0$

$\neq 0$ $= 0$

$\det((\omega+\lambda-1)D - \lambda\omega L - \omega U) = 0$

$0<\omega\leq 1$ 假设 $|\lambda|\geq 1$

~~$(\lambda-1)(\omega-1)\leq 0$~~

$|\omega+\lambda-1| \geq |\omega\lambda|$

且 $|\omega\lambda| \geq |\omega|$

~~$\det((\omega+\lambda-1)D - \lambda\omega L$~~

因为 $\lambda\omega(D-L-U)$ 对角占优

$\therefore$ $(\omega+\lambda-1)D-\lambda\omega L-\omega U$ 对角占优，矛盾

$\therefore$ $0\leq|\lambda|<1$ 即 $\rho(B_{SOR}^{\omega})<1$

4. $x^{k+1} = (I-\omega A)x^k + \omega b$

设 λ 为 $I-\omega A$ 特征值

$(I-\omega A)x = \lambda x$

$\lambda = \frac{x^T(I-\omega A)x}{x^Tx} = 1-\omega\frac{x^TAx}{x^Tx}$

$\therefore$ $0<\omega\frac{x^TAx}{x^Tx}<2$

$0<\omega<\frac{2}{\max(\frac{x^TAx}{x^Tx})}$ 而 $\max(\frac{x^TAx}{x^Tx}) = \rho(A)$

$\therefore$ $0<\omega<\frac{2}{\rho(A)}$

5. $F(x^{(k+1)}) = F(x^{(k)}+\alpha_k p^{(k)}) = \frac{1}{2}(Ax^{(k)}, x^{(k)}) - (b, x^{(k)}) + \frac{\alpha_k^2}{2}(Ap^{(k)}, p^{(k)})$

$+\alpha_k(Ax^{(k)}, p^{(k)}) - \alpha_k(b, p^{(k)})$

$= F(x^{(k)}) - \alpha_k(r^{(k)}, p^{(k)}) + \frac{\alpha_k^2}{2}(Ap^{(k)}, p^{(k)})$

$(\alpha_k = \frac{(r^{(k)}, p^{(k)})}{(Ap^{(k)}, p^{(k)})})$ $= F(x^{(k)}) - \frac{(r^{(k)}, p^{(k)})^2}{2(Ap^{(k)}, p^{(k)})} \leq F(x^{(k)})$

($r^{(k)}=0$ 时取 "=")

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6. (1) $V=(r, p, v_3, \cdots, v_n)$ $H=\begin{pmatrix} h_{11} & h_{12} & \cdots & h_{1n} \\ h_{21} & h_{22} & \cdots & h_{2n} \\ 0 & h_{32} & \cdots & h_{3n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & h_{nn} \end{pmatrix}$

$AV=VH$

$Ar=h_{11}r+h_{21}p$

$r^TA^2r=h_{11}r^TAr \qquad h_{11}=\dfrac{r^TA^2r}{r^TAr}$

$h_{21}=\|Ar-h_{11}r\|_2$

$p=\dfrac{(Ar-h_{11}r)}{\|Ar-h_{11}r\|_2}$ 其中 $h_{11}=\dfrac{r^TA^2r}{r^TAr}$

(2) $AV_1=V_1H_1+h_{21}v_2e_1^T$

$H_1y_1=\|r\|_2e_1$

$y_1=\dfrac{r^TAr}{r^TA^2r\,\|r\|_2}\|r\|_2H_1^{-1}e_1$

$z_1=\dfrac{r^TAr}{r^TA^2r\,\|r\|_2}\cdot\dfrac{1}{\|r\|_2}\cdot\dfrac{r^TAr}{r^TA^2r}r$

$x_1=x_0+z_1=x_0+\dfrac{1}{\|r\|_2}\cdot\dfrac{r^TAr}{r^TA^2r}r$

(3) $\|r_0-Az_m\|=h_{m+1,m}|e_m^Ty_m|$

只需关注 $h_{m+1,m}|e_m^Ty_m|$ 大小

感觉收敛，但证明暂未想出

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